THE FROBENIUS NUMBER OF DAO'S CYCLIC SEMIGROUPS

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ABSTRACT. For positive integers $a_1, \dots, a_n$ ($n \geq 2$, indices read cyclically) put $x_i = 1 + a_i + a_i a_{i+1} + \cdots + a_i a_{i+1} \cdots a_{i+n-2}$. In 2018 Hailong Dao asked whether the numerical semigroup $\langle x_1, \dots, x_n \rangle$, when its generators are coprime and minimal, has Frobenius number $(n-1)(a_1 a_2 \cdots a_n - 1)$, a direct extension of the two-generator formula. We prove this for every coprime list. The formula also follows from a theorem of Goto, Kien, Matsuoka and Truong once the semigroup ring is known to be cut out by the 2×2 minors of a cyclic $2 \times n$ matrix; that presentation has not been established for this family, and the proof here does not use it. The argument is elementary: one identity, $a_{i-1} x_i = x_{i-1} + N$ with $N = a_1 \cdots a_n - 1$, turns every representation into a mixed-radix numeral modulo N, and carrying digits describes each residue class of the semigroup exactly.

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1. INTRODUCTION

Two coprime integers $u, v \geq 2$ generate a numerical semigroup whose largest missing integer is $uv - u - v$. Write $u = a + 1$ and $v = b + 1$ and the formula becomes $ab - 1$: the Frobenius number is one less than the product of the shifts. Dao [5] asked on MathOverflow in 2018 whether this pattern survives with more generators, for a particular cyclic choice of generators that arose in his work on monomial curves [6]. The question had no answer for eight years. This note answers it.

A first case. Take $(a_1, a_2, a_3) = (1, 2, 3)$. The recipe below gives the generators

$$x_1 = 1 + 1 + 1 \cdot 2 = 4, \qquad x_2 = 1 + 2 + 2 \cdot 3 = 9, \qquad x_3 = 1 + 3 + 3 \cdot 1 = 7.$$

The integers not in $\langle 4, 7, 9 \rangle$ are $1, 2, 3, 5, 6, 10$, so the Frobenius number is 10. Dao's formula predicts $(n-1)(a_1 a_2 a_3 - 1) = 2 \cdot 5 = 10$.

In general, fix $n \geq 2$ and positive integers $a_1, \dots, a_n$, and read every index modulo n. Put

$$x_i = 1 + \sum_{j=0}^{n-2} a_i a_{i+1} \cdots a_{i+j} \qquad (1 \leq i \leq n),$$

so x_i is 1 plus the $n - 1$ products of consecutive a's that start at a_i and stop short of a full turn. Write $S = \langle x_1, \dots, x_n \rangle$ for the set of all sums $\sum \lambda_i x_i$ with $\lambda_i \in \mathbb{Z}_{\geq 0}$, and

$$P = a_1 a_2 \cdots a_n, \qquad N = P - 1.$$

When $n = 2$ the generators are $1 + a_1$ and $1 + a_2$, and $N = a_1 a_2 - 1$ is the classical answer.

Theorem 1. *If* $\gcd(x_1, \dots, x_n) = 1$*, then* $(n-1)N \notin S$ *and every integer* $m > (n-1)N$ *lies in* S*. So the Frobenius number of* S *is*

$$F(S) = (n-1)(a_1 a_2 \cdots a_n - 1).$$

Dao also assumed that the x_i generate S minimally. The proof does not need this. In every coprime case we computed (Remark 1) the generators were minimal anyway, so we do not claim the omission as a strengthening.

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Prior work. The family is the cyclic case of a class studied in commutative algebra. Identity (1) below says that the monomials X_{i-1} and $X_i^{a_{i-1}}$ differ in degree by the same amount N for every i, so the 2×2 minors of

$$M = \begin{pmatrix} X_2^{a_1} & X_3^{a_2} & \cdots & X_1^{a_n} \\ X_1 & X_2 & \cdots & X_n \end{pmatrix}$$

are homogeneous and vanish on the monomial curve $t \mapsto (t^{x_1}, \ldots, t^{x_n})$. Goto, Kien, Matsuoka and Truong [1, Theorem 1.2] proved that if a minimally generated numerical semigroup has its defining ideal equal to the ideal of 2×2 minors of a matrix of this shape, then its pseudo-Frobenius numbers are $\alpha, 2\alpha, \ldots, (n-1)\alpha$, where α is the common degree difference; Kien and Matsuoka [4, Theorem 3] extended this to cyclic two-row matrices with arbitrary exponents in both rows, where the progression starts at a shift h. For Dao's semigroups $\alpha = N$, and Theorem 1 would follow. What is missing is the presentation itself: we found no proof that the minors of M generate the whole defining ideal for this family, and recent work on when such presentations hold [7, 3] reduces the question to numerical conditions without settling this case. For $n = 3$, Herzog's theorem [2] gives a presentation by the 2×2 minors of some 2×3 matrix whenever the semigroup is not symmetric, but not necessarily by M. The proof below avoids the presentation and uses only arithmetic.

2. The carry identity

Multiply the sum defining x_i by a_{i-1}. Each product $a_i a_{i+1} \cdots a_{i+j}$ becomes $a_{i-1} a_i \cdots a_{i+j}$, which is a term of x_{i-1}, and the last one becomes the full product P. The constant 1 becomes a_{i-1}, also a term of x_{i-1}. Only the constant 1 of x_{i-1} is not produced, so

$$a_{i-1}\, x_i = x_{i-1} + N \qquad (1 \le i \le n). \tag{1}$$

In words: a_{i-1} copies of x_i are worth exactly one copy of x_{i-1} plus N. In the first case, $1 \cdot 9 = 4 + 5$, $2 \cdot 7 = 9 + 5$ and $3 \cdot 4 = 7 + 5$.

Summing (1) over i, the x_{i-1} on the right run over all generators once, giving

$$\sum_{i=1}^{n} (a_{i-1} - 1)\, x_i = nN. \tag{2}$$

So the vector with $a_{i-1}-1$ copies of each x_i is worth n times N; for $(1, 2, 3)$ it is $0 \cdot 9 + 1 \cdot 7 + 2 \cdot 4 = 15$.

Two consequences. If $P = 1$ then every $a_i = 1$ and every $x_i = n \ge 2$, which contradicts coprimality, so $N \ge 1$. And by (1) a common divisor of x_n and N divides $x_{n-1} = a_{n-1} x_n - N$, then x_{n-2}, and so on round to x_1. Hence

$$\gcd(x_n, N) = 1. \tag{3}$$

3. Digit vectors

Read (1) modulo N: it says $x_{i-1} \equiv a_{i-1} x_i$, and iterating from x_n,

$$x_i \equiv \pi_i\, x_n \pmod{N}, \qquad \pi_i = a_i a_{i+1} \cdots a_{n-1}\ (1 \le i < n), \quad \pi_n = 1. \tag{4}$$

So modulo N each generator is a fixed multiple of x_n, and a sum $\sum d_i x_i$ is x_n times the number $\sum d_i \pi_i$. That number looks like a numeral with place values π_i, and it is one when each digit stays below its radix. This motivates the following definition. Call $d = (d_1, \ldots, d_n)$ a *digit vector* if $0 \le d_i \le a_{i-1} - 1$ for each i, and write $w(d) = \sum d_i x_i$ for its value.

So $w(d) \equiv x_n \sum_i d_i \pi_i$, and $\sum_i d_i \pi_i$ is an ordinary mixed-radix numeral: its place values are $\pi_n = 1, \pi_{n-1} = a_{n-1}, \pi_{n-2} = a_{n-2} a_{n-1}, \ldots, \pi_1$, and the digit in place π_i runs below a_{i-1} (the place π_1 has radix a_n, and $\pi_1 a_n = P$). The standard facts about such numerals give the following lemma.

Lemma 1. *The numbers $\sum_i d_i \pi_i$, as d runs over digit vectors, are exactly $0, 1, \ldots, N$, each once. The value 0 comes from $d = 0$ and the value N from the maximal vector $d_i = a_{i-1} - 1$.*

Proof. There are $\prod a_{i-1} = P = N + 1$ digit vectors. Every integer $0 \le t \le N$ has an expansion: divide by the radix of the lowest place, keep the remainder as the lowest digit, and repeat with the quotient. Two vectors with the same numeral agree in the lowest digit (both equal the remainder), and then in every digit by induction. The maximal vector gives $\sum(a_{i-1} - 1)\pi_i = P - 1$ because the sum telescopes. □

Combining the lemma with (3): multiplication by x_n permutes the residues modulo N, so the values $w(d)$ hit every nonzero residue class modulo N exactly once, and the zero class twice, at $w(0) = 0$ and at $w(\max) = nN$ by (2). Every digit vector has $w(d) \le nN$, again by (2). For $(1, 2, 3)$ the six digit vectors have values $0, 4, 8, 7, 11, 15$, with residues $0, 4, 3, 2, 1, 0$ modulo 5.

4. Proof of Theorem 1

The plan is to describe, for each residue r modulo N, the set $S \cap (r + N\mathbb{Z})$. It turns out to be a ladder: a lowest rung at the digit value $w(d)$ in that class, then every rung N apart above it. Moving down the ladder is carrying, moving up is uncarrying, and the Frobenius number is the highest missing rung, read off from (2). Figure 1 shows the ladders for $(1, 2, 3)$.

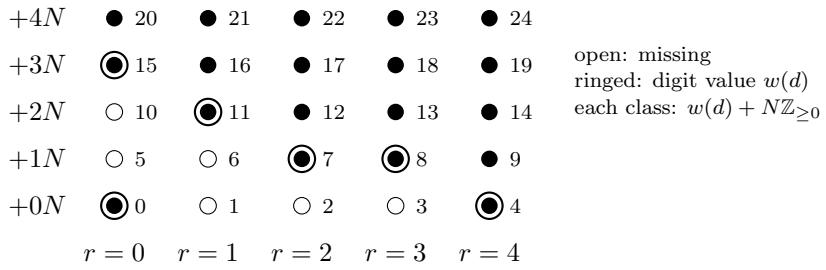


Figure 1. The residue classes modulo $N = 5$ of $S = \langle 4, 9, 7\rangle$, from $(a_1, a_2, a_3) = (1, 2, 3)$. Each column is a class; filled dots are elements of S. In each nonzero class the lowest element is the value of the unique digit vector in that class, and everything N above it is in S. In the zero class the digit values are 0 and $nN = 15$, so 5 and 10 are missing and $10 = (n - 1)N$ is the Frobenius number.

Carrying down. Let $s = \sum \lambda_i x_i$ with $\lambda \ne 0$. If some $\lambda_i \ge a_{i-1}$, replace a_{i-1} copies of x_i by one copy of x_{i-1}. By (1) the value drops by exactly N, the residue modulo N is unchanged, and the vector stays nonzero, since a copy of x_{i-1} was added. The value cannot drop forever, so the process stops at a nonzero digit vector d with

$$s = w(d) + kN \qquad \text{for some } k \ge 0. \tag{5}$$

Carrying up. Conversely, if $\lambda \ne 0$ then some $\lambda_{i-1} \ge 1$, and replacing one copy of x_{i-1} by a_{i-1} copies of x_i raises the value by exactly N and keeps the vector nonzero. Repeating, $w(d) + kN \in S$ for every nonzero digit vector d and every $k \ge 0$.

Together with Lemma 1 and (3) this gives the ladders. For a nonzero residue r let $d(r)$ be the unique digit vector with $w(d(r)) \equiv r$; then

$$S \cap (r + N\mathbb{Z}) = \{\, w(d(r)) + kN : \ k \ge 0 \,\}.$$

In the zero class, a nonzero element of S carries down to a nonzero digit vector with residue 0, which must be the maximal vector, so

$$S \cap N\mathbb{Z} = \{0\} \cup \{\, nN + kN : \ k \ge 0 \,\}.$$

The first display reads: in class r the members of S are the digit value $w(d(r))$ and everything N above it. The second reads: the multiples of N in S are 0 and those from nN upwards.

Now the Frobenius number. In a nonzero class the largest missing integer is $w(d(r)) - N$, and $w(d(r)) < nN$ because the value nN belongs to the zero class; so it is at most $(n-1)N-1$.

In the zero class the largest missing integer is $(n-1)N$, which is positive because $n \geq 2$ and $N \geq 1$. Hence $(n-1)N \notin S$ and every larger integer is in S. □

5. Remarks and checks

Remark 1. Dao's question assumed the x_i generate S minimally, and the proof above never uses it. We tested whether it can fail: in all 3,703 coprime lists with $2 \leq n \leq 5$ and entries at most 8 (for $n \leq 3$) or 5 (for $n = 4, 5$), and 14,459 random coprime lists with $n \leq 7$, no generator was a combination of the others. We have not proved that minimality always holds.

Remark 2. The theorem of Goto, Kien, Matsuoka and Truong predicts more than the Frobenius number: if the presentation by the minors of M holds, the pseudo-Frobenius numbers of S (the missing f with $f + x_i \in S$ for every i) are exactly $N, 2N, \ldots, (n-1)N$. We computed them in 2,186 random coprime cases with $n \leq 5$ and found exactly this set every time. The ladders above suggest a direct proof, which we have not written down.

Verification. The accompanying repository checks Theorem 1 and the ladder description of every residue class against a brute-force computation for all 2,379 coprime lists with $2 \leq n \leq 6$ and entries at most $8, 8, 5, 4, 3$ respectively. An independent program, sharing no code with the first, recomputes the same Frobenius numbers by shortest paths on residues modulo the smallest generator. Further scripts reproduce the worked example and the counts in the remarks. The whole proof of Theorem 1, including Lemma 1 and the carrying arguments, is formalised in Lean 4 with Mathlib, following the steps above.

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